Iodine Brachytherapy as an Alternative to Enucleation for Large Uveal Melanomas

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Purpose: To evaluate the safety and efficacy of iodine 125 plaque brachytherapy (IBT) for large uveal melanomas.

Design: Retrospective, nonrandomized comparative trial (historical control).

Participants: One hundred twenty-one consecutive patients with a large uveal melanoma according to the Collaborative Ocular Melanoma Study (COMS) criteria who attended a national ocular oncology service.

Methods: Ninety-seven patients (80%) underwent primary IBT (mean dose to tumor apex, 87 Gy) with noncollimated 20- to 25-mm plaques. Assessment of metastatic disease at death and visual outcome followed COMS guidelines. Time to low vision (20/70 or worse) and blindness (loss of 20/400 vision) in the study eye were modeled by Cox proportional hazards regression, based on both single- and repeated-failure data sets. Person-years of retained vision were calculated.

Main Outcome Measures: All-cause and melanoma-specific survival, local and distant recurrence, and preservation of vision and cosmesis.

Results: Median tumor height was 10.7 mm (range, 4.5–16.8 mm), and largest basal tumor diameter was 16.1 mm (range, 7.3–25.0 mm). The Kaplan-Meier estimate for all-cause and melanoma-specific survival was 62% (95% confidence interval [CI], 49%–72%) and 65% (95% CI, 52%–75%) at 5 years. The corresponding estimate for local tumor recurrence was 6% (95% CI, 2%–14%) and for major cosmetic abnormality was 38% (95% CI, 26%–52%). The median visual acuity in the study eye was 20/100 at baseline and 20/1600 at 2 years after treatment. The Kaplan-Meier estimate for avoiding low vision and blindness was 11% (95% CI, 4%–24%) and 26% (95% CI, 16%–37%) at 2 years, respectively. Tumor height and location entirely posterior to the ora serrata were the most robust predictors of visual loss. In this series, 49 person-years without low vision (median, 0.6 years; range, 0.04–8.2 years) and 111 person-years without blindness (median, 1.0 years; range, 0.03–8.6 years) in the treated eye were conserved.

Conclusions: Iodine 125 plaque brachytherapy seems to be a safe and effective alternative to enucleation with regard to survival and local tumor control. It provides a fair chance of preserving the eye with acceptable cosmesis and a reasonable chance of conserving useful vision for 1 to 2 years. *Ophthalmology 2003;110: 2223–2234 © 2003 by the American Academy of Ophthalmology.*

Most large malignant melanomas of the uvea are managed with primary enucleation.[1] This practice is based on tradition and on the frequent opinion that enucleation is the treatment of choice when there is little hope for regaining or saving useful vision, as is often the case when the tumor is large.[2,3] When the ocular oncologist believes that reasonable hope for saving vision exists, however, several options are available.[4,5] These include charged particle therapy,[6,7] fractionated stereotactic radiotherapy,[8] gamma-knife radiosurgery,[9] and plaque radiotherapy.[5,10] Local resection can also be considered for selected large uveal melanomas.[11]

It is reasonable to evaluate alternative treatments for large uveal melanomas, because tumor size is the strongest clinical indicator of high risk of metastatic death[12] and it is estimated that micrometastases typically develop within 5 years before the diagnosis of the primary tumor.[13,14] Intuitively, enucleation of a large uveal melanoma thus offers only limited survival benefit, and preliminary data from the prospective, randomized Collaborative Ocular Melanoma Study (COMS) suggest this to be true even when the tumor is medium sized.[15]

None of the alternative therapies has so far been evaluated specifically as the main treatment for large uveal melanoma. The COMS considered only enucleation for patients who had a large melanoma,[1,16] whereas those with a medium melanoma were randomized between enucleation and

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iodine plaque radiotherapy, which may save vision.[15,17] Referring ophthalmologists and ocular oncologists alike, when confronted with a large uveal melanoma, consequently may have difficulty in judging exactly when and to what extent there may be hope for saving vision.

We report here the population-based data of 121 consecutive patients, of whom 97 had a large choroidal melanoma managed with primary iodine 125 (I^{125}) plaque brachytherapy (IBT), a method that, during the last decade, has become the most frequent treatment for this type of tumor in our center.

Materials and Methods

Aims of the Study

The primary aim was to assess survival and local tumor control in a population-based, consecutive group of patients who underwent primary I^{125} episcleral IBT for a uveal melanoma that was large by the COMS criteria. A secondary goal was to evaluate its efficacy in retaining vision and cosmesis.

Inclusion Criteria

Eligible for this study were patients who were diagnosed with a uveal melanoma that met the COMS criteria of large tumor (largest basal diameter [LBD] >16.0 mm and height >2.0 mm, height >10.0 mm regardless of LBD, or a peripapillary tumor >8.0 mm by height and located <2.0 mm from the optic disc)[1] that was managed with IBT between November 1, 1990, and June 1, 2001. The LBD and tumor location were determined primarily by indirect ophtalmoscopy, B-scan ultrasonography, and, in selected cases, computed tomography (CT) or magnetic resonance imaging (MRI). Tumor height was based on A- and B-scan ultrasonography.

Analogous to COMS, the patient was ineligible if the tumor involved the iris; if more than 50% of it was located in the ciliary body; if it was a diffuse, ring, or multifocal type; or if it had an extrascleral extension of 2.0 mm or more. Also ineligible were patients who had undergone other treatment for their uveal melanoma before IBT. To enhance generalizability, we did not consider patients ineligible if they had a history of other coexisting disease that threatened survival, another cancer, or metastatic melanoma at the time of diagnosis, provided that treatment was otherwise indicated.

The files of the Ocular Oncology Service, Department of Ophthalmology, Helsinki University Central Hospital, a tertiary referral center that managed 90% of uveal melanomas in Finland during the study period, were reviewed. A total of 121 patients with a large uveal melanoma were identified, of whom 24 were managed by methods other than IBT: 12 with cobalt and ruthenium brachytherapy at a time when indications for IBT were evolving, one with proton beams (in Clatterbridge, UK), and 11 with primary enucleation. One patient requested primary enucleation; four tumors could not be covered with the largest brachytherapy plaque; three had caused an extensive retinal detachment and iris neovascularization, which indicated poor prognosis for saving the eye; and two had caused a hemorrhagic choroidal detachment with or without subretinal hemorrhage, which made it impossible to place the plaque accurately and also greatly worsened the prognosis for saving the eye. One elderly patient initially declined treatment, and the eye was enucleated after the tumor had extended outside the globe.

Consequently, 97 of the 121 patients (80%) underwent IBT as primary treatment for a large uveal melanoma and were enrolled. One of them had extrascleral extension noted at the time of plaque insertion, and enucleation was performed instead. This patient was included in all analyses in accordance with the intention-to-treat principle.

Clinical Evaluation

The diagnosis of uveal melanoma was based on clinical, ultrasonographic, and, in selected cases, fluorescein angiographic findings. The clinical parameters recorded at baseline included best-corrected visual acuity (VA) and intraocular pressure. The VA was measured using a test type projector (Rodavist 2 and 524; Rodenstock GmbH, Ottobrunn, Germany) and a scale based on even decimals from 1.0 (20/20) to 0.1 (20/200) plus 0.05 (20/400). A VA worse than 20/400 was recorded as counting fingers at 2 m, 1 m, and 0.5 m, hand movement, light perception, and no light perception. Biomicroscopy and indirect ophthalmoscopy were performed to localize the tumor margins; to evaluate its growth pattern, exudative retinal detachment, and vitreous hemorrhage; and to estimate distance from the posterior tumor margin to the optic disc and foveola. Gonioscopy was used to determine the anterior border of the tumor and to detect neovascularization of the iris, chamber angle, or both.

Liver function tests (aspartamine aminotransferase, alanine aminotransferase, alkaline phosphatase, and lactate dehydrogenase), a chest radiogram, and an abdominal ultrasonography were obtained to detect metastases. In case of amelanotic and otherwise atypical tumors, mammography and whole body CT scans were ordered to exclude intraocular metastasis and a primary malignancy elsewhere. One patient had metastases from uveal melanoma at baseline and received IBT as a palliative treatment.

Elements of Informed Consent

The first patients in this study were managed with IBT because they declined primary enucleation. Encouraged by the outcome, patients who had a large uveal melanoma diagnosed were subsequently informed, in accordance with the Act of Patients' Rights (1992/785), that two treatments were available: primary enucleation and IBT, followed by secondary enucleation should a local recurrence or intractable complications from irradiation develop. They then were informed that local recurrence after enucleation was unlikely, whereas local recurrence and secondary enucleation were expected at least 10% and 20% of the time after IBT, respectively, based on data for medium melanomas,[1,18–20] and that neither treatment was expected to give any significant advantage in preventing metastasis and improving survival as compared with the other.

Enucleation was recommended if the patient had a ring melanoma, a melanoma larger than the largest brachytherapy plaque, a tumor that grew around the optic disc, a tumor that was associated with notable suprachoroidal bleeding or extensive retinal detachment, vitreous hemorrhage, or iris neovascularization. Otherwise, the patients were managed according to their personal preference after informed consent was obtained.

If a local recurrence developed, secondary enucleation was recommended; patients who declined were managed with reirradiation. If visibility to the fundus was lost, the patient was also offered secondary enucleation as a safety measure. Most patients elected to continue follow-up with ultrasonography, CT, and MRI.

The study followed the tenets of the Declaration of Helsinki.

Iodine Brachytherapy

Iodine 125 brachytherapy is a suitable technique for treatment of large intraocular tumors because of its low gamma energy (27–35 keV) and practical half life (59.4 days). The applicators were crafted in 1991 by a local goldsmith to conform with ruthenium 106 plaques already in use.[21] Four 0.5-mm thick unrimmed plaques were used to irradiate large melanomas: CCB (diameter 20 mm, circular), CCC (25 mm, circular), COB (20 mm, notch for the optic nerve), and CIB (diameter 20 mm, notch for the limbus). The I^{125} seeds were attached with silicone rubber (RTV 3140; Dow Corning Corporation, Midland, MI), which increased the plaque thickness to 1.0 to 1.5 mm. The number of seeds (555 MBq; model 6702; Mediphysics Inc., Arlington Heights, IL) ranged from 8 to 12. The dose distribution for each combination of seeds and plaque was calculated with commercial brachytherapy software (Cadplan; Varian Dosetek, Helsinki, Finland). The angular and distance dependence dose rates were determined from tabulated correction matrices that included absorption and scatter correlations.[21–23] In 1995, the software was modified according to the American Association of Physicists in Medicine Radiation Therapy Committee Task Group No. 43.[24]

The prescription point was tumor height plus 1 mm for the sclera. The prescription dose was initially 100 Gy to the apex of the tumor, and 80 Gy if the tumor was very thick, in an effort to limit complications. Since 1997, the prescription dose has been 80 Gy to the apex as a rule, and very thick tumors have received a prescription dose of 70 to 60 Gy. Treatment time and dose rate were calculated from the central axis depth dose curve. Seeds were replaced when treatment times approached 2 weeks. The median prescription point was 11.3 mm from the scleral surface, median prescription dose to the apex was 87 Gy (range, 42–109 Gy), and median dose rate at the tumor apex was 0.57 Gy/h (range, 0.13–2.17 Gy/h). In practice, 90% of patients received a dose between 65 Gy and 102 Gy, and 69% received a dose between 80 Gy and 100 Gy.

A minimum safety margin of 2 mm around the tumor was desirable, but was not an absolute requirement. Tumors close to the optic disc and macula often were irradiated with a smaller or no safety margin toward these structures.

Assessment of Local Tumor Control

Patients were prospectively followed up at 3, 6, and 12 months after brachytherapy and twice yearly thereafter for 3 years in Helsinki, and then at least once yearly, often in a regional hospital. Biomicroscopy, diascleral transillumination, indirect ophthalmoscopy, B-scan ultrasonography, and, in selected cases, CT and MRI were used to assess tumor control. Recurrence was coded as vertical, marginal, ring melanoma, or extrascleral.[25] An unequivocal change of 1.0 mm or more in either dimension indicated local marginal or vertical recurrence, respectively.

Assessment of Cosmesis

Cosmetic outcome was classified as normal, minor abnormality (e.g., persistent redness, inconspicuous scleral thinning, slightly asymmetric pupil, or cataract visible to the naked eye without other cosmetic abnormality) and major abnormality (e.g., visible corneal opacity, rubeosis, or hyphaema; phthisis bulbi; or presence of two or more minor abnormalities). Redness resulting from recent surgery and similar temporary abnormalities not present at the next examination were not counted.

Assessment of Metastatic Status

The liver function tests and upper abdominal ultrasonography were repeated annually. In selected cases, they were also performed at 6 and 18 months after IBT.

Survival status was classified as alive with or without metastasis; dead of metastatic melanoma; dead of other cancer and nonneoplastic causes, and unknown. Death certificates were obtained from Statistics Finland, and charts relating to terminal illness were retrieved. The cause of death was coded as diagnostic of melanoma metastasis if histopathologic diagnosis was confirmed in our laboratory by immunohistochemistry or, if the specimen was unavailable, if the original histopathologic report documented unequivocal melanin. Otherwise, the cause of death was coded consistent with metastatic melanoma if based on histopathologic analysis. Patients without histopathologic analysis who had evidence of metastasis and a progressive course with no evidence of a second cancer were considered to have suspected melanoma metastasis.[26]

Death resulting from other cancer necessitated histopathologic diagnosis. Death resulting from a nonneoplastic cause was classified as certain only if an autopsy had been performed. Otherwise, we required that the patient had no evidence of metastatic disease at last review, performed no more than 6 months before death.

Statistical Methods

Follow-up data up to the date of analysis on November 30, 2001, were collected prospectively into a dedicated database that was restructured in January 2001 to allow analysis of repeated events (Access 97; Microsoft Corporation, Seattle, WA).

The data were analyzed with Stata statistical software (Release 7.0; Stata Co., College Station, TX). Primary endpoints were all-cause and melanoma-related survival, time to local recurrence, and time to development of metastases, assessed by Kaplan-Meier analysis.[27] To address selection bias, we also considered the survival of all 121 patients who had a large uveal melanoma. Kaplan-Meier analysis was also used to analyze time to secondary enucleation and cosmetic outcome. Percentage regression in tumor height was plotted at 6-month intervals.

Mean VA, after transformation to a logarithmic scale and stratified by factors associated with poor vision outcomes, was plotted at 6-month intervals according to the COMS Medium Tumor Study.[17] Measurements below 20/1600, such as counting fingers or hand movement, were set equal to 20/1600. Enucleated eyes and eyes with no light perception were counted as having 20/2500 vision.[17] If the patient did not have a measurement available at all appropriate intervals, the VA was taken to be the average of the previous and the next visits.[17] If there was a longer gap in follow-up, the patient was censored from the analysis.

Loss of VA levels 20/40, 20/65, 20/200 and loss of more than 6 lines of vision, corresponding to quadrupling of the minimal angle of resolution from the baseline, were analyzed with the Kaplan-Meier method according to the COMS Medium Tumor Study. Loss of vision was recorded only when confirmed at the next follow-up examination.[17] In addition, development of unilateral low vision (VA 20/70 or worse) and unilateral blindness in the study eye (VA less than 20/400) according to the World Health Organization (WHO) criteria were analyzed.

Factors predicting low vision and blindness were modeled by univariate and multivariate Cox proportional hazards regression, based on both single-failure (time to first event) and multiple-failure (repeated events) data sets.[28,29] Standard errors were calculated using the robust variance estimator of Lin and Wei,[30] and tied survival times were handled with Efron approximation.[27] Age at diagnosis, LBD, tumor height, and the distance of posterior tumor margin to the foveal avascular zone (FAZ) were modeled as continuous variables. Location of the anterior tumor margin was dichotomized (behind the ora serrata, in the ciliary body). Bivariate models were used to isolate the most robust independent predictors of visual loss. The number of variables in the final model was restricted, based on a rule to have at least 15 to 20 events per each additional variable.[31] The regression coefficients and hazard ratios (HR) with 95% confidence intervals (CIs) were calculated.

We also estimated person-years of vision conserved in the study eye by counting the total time at risk for the COMS and WHO adverse visual outcomes, allowing for recovery and repeated loss of vision with the multiple failures per subject analysis.

Results

The median age at diagnosis of the 97 enrolled patients (male: female, 49:48), all white persons, was 64 years (range, 24–82 years). The median follow-up time was 3.6 years (range, 0.3–10.5 years).

Baseline Tumor Characteristics

The median tumor height and LBD were 10.7 mm (range, 4.5–16.8 mm) and 16.1 mm (range, 7.3–25.0 mm), respectively (Fig 1A; Table 1). Of the 97 tumors, 10 (10%) were classified as large based on their peripapillary location,[1] and 61 (63%) involved the ciliary body (Fig 1B; Table 1). The posterior margin of the tumor touched the optic disc in 15 eyes (15%) and extended to within 2 mm of it in 8 eyes (8%). The median distance to the center of the FAZ was 6.0 mm (range, 0–18 mm).

Tumor shape was collar button or peaked in 82 eyes (85%), and dome or flat in 10 eyes (10%; Table 1). The intraocular pressure was normal in most eyes (94%). Exudative retinal detachment was present in 83 eyes (86%) and vitreous hemorrhage was present in 13 eyes (13%).

Only three patients had 20/20 vision at diagnosis, whereas 29 (30%) had an acuity of 20/200 or worse (Table 1). According to the WHO definitions, 35 study eyes (36%) had low vision and 19 (19%) were blind.

As compared with the COMS Large Tumor Study (Table 1),[1] the tumor height was skewed toward higher values (mean, 10.8 vs. 9.5 mm; $P<0.001$, Kruskal-Wallis test) and the LBD toward smaller values (mean, 16.2 vs. 17.2 mm; $P = 0.001$, Kruskal-Wallis test). Correspondingly, tumor shape in the present trial was more frequently collar button or peaked (85% vs. 60%; Table 1; $P<0.001$, Pearson's chi-square test).

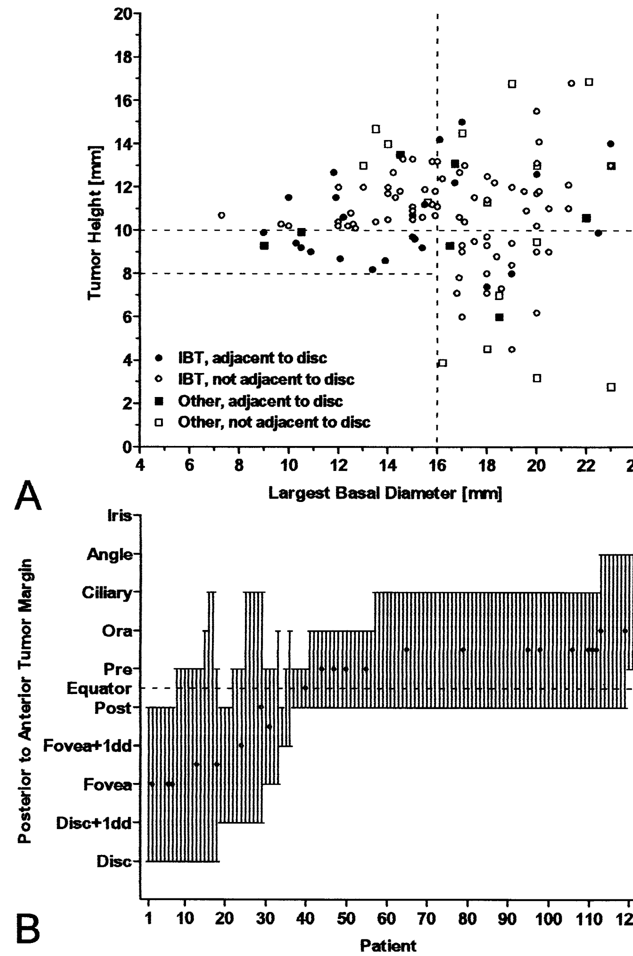

Figure 1. Characteristics of 121 large uveal melanomas, of which 97 were managed with iodine brachytherapy (IBT). **A,** Scatterplot of largest basal tumor diameter against tumor height. Dashed lines indicate Collaborative Ocular Melanoma Study (COMS) criteria for large melanoma. Circles indicate tumors managed with IBT, and boxes indicate tumors managed with other options. Open symbols indicate tumors located 2 mm or further from the optic disc, and solid symbols indicate those located within 2 mm from it. **B,** Plot of tumor location. Bars indicate the position of the anterior and posterior tumor margin. Bars with circles indicate tumors that were not managed with IBT.

Characteristics of Patients Not Managed by Iodine 125 Brachytherapy

The median age of the 24 patients treated with a method other than IBT was 62 years (range, 17–84 years; Table 1). The tumor dimensions were comparable, but tumor shape was more often dome or flat ($P = 0.03$, Pearson's chi-square test), and visual acuity of the tumor eye was significantly worse (Table 1; $P = 0.035$, Kruskal-Wallis test) as compared with eyes treated with IBT.

Survival

Of the 97 patients, 34 died during the follow-up and 28 of these deaths were classified as being the result of metastatic uveal melanoma. Of the 28 melanoma deaths, 19 (68%) were histopathologically confirmed (17 were diagnostic of and 2 consistent with metastatic melanoma), and 9 were suspected on the basis of clinical data. An additional 8 patients were alive a median of 14 months (range, 2–32 months) after diagnosis of metastases. One patient was dead of histopathologically confirmed metastatic colonic carcinoma, and 5 patients died without evidence of melanoma metastases (3 of myocardial infarction and 2 of cerebrovascular accident).

The Kaplan-Meier estimate for all-cause survival was 62% at 5 years (95% CI, 49%–72%) and 48% at 8 years (95% CI, 33%–61%; Fig 2A). The estimated melanoma-specific survival was 65% (95% CI, 52%–75%) and 54% (95% CI, 38%–68%) at 5 and 8 years, respectively (Fig 2B). These survival curves are visually comparable with the melanoma-specific survival after enucleation in the COMS Large Tumor Study (based on the originally published graph,[16] which corresponded to melanoma-specific mortality defined as death with histopathologically confirmed metastasis

Table 1. Characteristics of 121 Patients with a Large Uveal Melanoma by the Collaborative Ocular Melanoma Study (COMS) Criteria by Method of Treatment, Compared with COMS Data*

	Iodine Brachytherapy n = 97, n (%)	Other Treatment n = 24, n (%)	*P*	All Large Melanomas n = 121, n (%)	Collaborative Ocular Melanoma Study Data n = 1003, n (%)	*P*†
Age (yrs)			0.77‡			0.30‡
<40	7 (7)	3 (13)		10 (8)	90 (9)	
40–49	11 (11)	3 (13)		14 (12)	135 (13)	
50–59	19 (20)	2 (8)		21 (17)	210 (21)	
60–69	30 (31)	8 (33)		38 (31)	286 (29)	
70–79	23 (24)	4 (17)		27 (22)	228 (23)	
>80	7 (7)	4 (17)		11 (9)	54 (5)	
Gender			0.18§			0.20§
Male	49 (51)	16 (67)		65 (54)	576 (57)	
Female	48 (49)	8 (33)		56 (46)	427 (43)	
Race			N/A			0.11‡
White, non-Hispanic	97 (100)	24 (100)		121 (100)	970 (97)	
Other	0 (0)	0 (0)		0 (0)	33 (3)	
Apical height, mm			0.83‡			<0.001‡
≤5.0	1 (1)	4 (17)		5 (4)	90 (9)	
5.1–7.0	2 (2)	2 (8)		4 (3)	109 (11)	
7.1–9.0	15 (15)	1 (4)		16 (13)	230 (23)	
9.1–11.0	37 (38)	5 (21)		42 (35)	281 (28)	
11.1–13.0	31 (32)	5 (21)		36 (30)	191 (19)	
13.1–15.0	9 (9)	5 (21)		14 (12)	73 (7)	
>15.0	2 (2)	2 (8)		4 (3)	28 (3)	
Not reported	0 (0)	0 (0)		0 (0)	1 (0)	
Largest basal diameter, mm			0.12‡			0.001‡
≤13.0	20 (21)	2 (8)		22 (18)	78 (8)	
13.0–14.9	11 (11)	3 (13)		14 (12)	82 (8)	
15.0–16.9	24 (25)	4 (17)		28 (23)	215 (22)	
17.0–18.9	17 (18)	6 (25)		23 (19)	334 (34)	
19.0–20.9	18 (19)	4 (17)		22 (18)	183 (18)	
21.0–22.9	4 (4)	2 (8)		6 (5)	73 (7)	
≥23.0	3 (3)	2 (8)		5 (4)	30 (3)	
Not reported	0 (0)	1 (4)		1 (1)	8 (1)	
Anterior border			0.44‡			0.024‡
Postequator	8 (8)	3 (13)		11 (9)	125 (12)	
Ora equator	28 (29)	9 (38)		37 (31)	357 (36)	
Ciliary body	54 (56)	9 (38)		63 (52)	441 (44)	
Chamber angle	7 (7)	3 (13)		10 (8)	57 (6)	
Indeterminate	0 (0)	0 (0)		0 (0)	23 (2)	
Proximity to optic disc, mm			0.29‡			0.24‡
≥2.0	73 (75)	15 (63)		88 (73)	676 (67)	
<2.0	23 (24)	8 (33)		31 (26)	290 (29)	
Not reported	1 (1)	1 (4)		2 (2)	37 (4)	
Distance to the center of FAZ, mm‖			0.78‡			N/A‡
0.0–2.0	15 (15)	4 (17)		19 (16)	82 (13)	
2.1–4.0	13 (13)	4 (17)		17 (14)	155 (25)	
4.1–6.0	23 (24)	4 (17)		27 (22)	183 (29)	
6.1–8.0	12 (12)	1 (4)		13 (11)	102 (16)	
>8.0	33 (34)	8 (33)		41 (34)	95 (15)	
Not reported	1 (1)	3 (13)		4 (3)	6 (1)	
Shape			0.035¶			<0.001¶
Dome or flat	10 (10)	5 (21)		15 (12)	392 (39)	
Collar button, peaked or irregular	82 (85)	15 (63)		97 (80)	601 (60)	
Indeterminate	5 (5)	4 (17)		9 (7)	10 (1)	
Retinal detachment			0.090¶			0.46¶
None noted	12 (12)	3 (13)		15 (12)	172 (17)	
Present, exudative	83 (86)	18 (75)		101 (83)	807 (80)	
Indeterminate	2 (2)	3 (13)		5 (4)	24 (2)	
Vitreous hemorrhage			1.0§			0.051§
Present	13 (13)	3 (13)		16 (13)	76 (8)	
Visual acuity, tumor eye			0.019‡			0.24‡
≥20/20	3 (3)	1 (4)		4 (3)	123 (12)	
20/25–20/32	12 (12)	2 (8)		14 (12)	136 (14)	
20/40–20/80	28 (29)	3 (13)		31 (26)	265 (26)	
20/100–20/160	25 (26)	3 (13)		28 (23)	108 (11)	
≤20/200	29 (30)	15 (63)		44 (36)	360 (36)	
Not reported	0 (0)	0 (0)		0 (0)	11 (1)	

Table 1. Continued

	Iodine Brachytherapy n = 97, n (%)	Other Treatment n = 24, n (%)	P	All Large Melanomas n = 121, n (%)	Collaborative Ocular Melanoma Study Data n = 1003, n (%)	P†
IOP in tumor eye, mmHg			0.42‡			1.0‡
<20	91 (94)	20 (83)		111 (92)	842 (84)	
20–23	3 (3)	0 (0)		3 (2)	3 (0)	
>23	2 (2)	2 (8)		4 (3)	38 (4)	
Not reported	1 (1)	2 (8)		3 (2)	120 (12)	

FAZ = foveal avascular zone; IBT = iodine brachytherapy; IOP = intraocular pressure; N/A = not applicable.
*The Collaborative Ocular Melanoma Study Group. The Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation of large choroidal melanoma I: characteristics of patients enrolled and not enrolled. COMS report no. 9. Am J Ophthalmol 1998;125:767–78.
†IBT compared with COMS data.
‡Kruskal-Wallis test, two-sided (excluding "not reported").
§Fisher exact test, two-sided.
‖Data taken from the COMS Medium Tumor Study (Melia BM, Abramson DH, Albert DM, et al. Collaborative Ocular Melanoma Study (COMS) randomized trial of I-125 brachytherapy for medium choroidal melanoma. I. Visual acuity after 3 years. COMS report no. 16. Ophthalmology 2001;108:348–66).
¶Pearson chi-square test, two-sided.

or, if histopathologic analysis was not available to confirm the source of metastasis, with suspected melanoma metastasis; Diener-West M, personal communication, 2002).

Tumor Regression and Local Recurrence

After IBT, tumor height regressed a median of 38% (range, 85% [14.3 mm] decrease to 5% [0.5 mm] increase) by 1 year and 48% (range, 82% [10.7 mm] decrease to 4% [0.4 mm] increase) by 3 years (Fig 2C). By 1 and 3 years, 30% and 23% of tumors had regressed by less than one third, and 14% and 23% had regressed by two thirds or more in height, respectively.

A local recurrence was diagnosed in 5 patients, all within 3 years from treatment. Two were classified as extrascleral marginal (perilimbal at tumor margin), 2 as vertical, and 1 as a ring melanoma. Two were controlled with additional IBT (1 later experienced an extrascleral marginal recurrence), and 3 were managed with secondary enucleation. The Kaplan-Meier estimate for developing first local recurrence was 6% (95% CI, 2%–14%) by 5 years (Fig 2D).

Enucleation and Cosmetic Outcome

In addition to the 3 eyes removed after recurrences, one primary enucleation instead of IBT was performed when extraocular extension was detected at surgery, and 8 eyes were removed a median of 12.5 months (range, 0.3–58 months) after diagnosis because of complications, mostly uncontrolled neovascular glaucoma. The Kaplan-Meier estimate for conservation of the eye was 84% (95% CI, 72%–92%) at 5 years (Fig 2E).

Of the retained eyes, 41 had normal cosmesis at last follow-up. The Kaplan-Meier estimate of a minor cosmetic abnormality was 66% (95% CI, 53%–79%) at 5 years from treatment. The corresponding estimate for a major cosmetic change was 38% (95% CI, 26%–52%).

Preservation of Visual Acuity

The median VA in the study eye was 20/100 at baseline, 20/1600 (counting fingers to hand movements) at 1 and 2 years, and 20/2500 (light perception) at 3 years from diagnosis.

The Kaplan-Meier estimate for retaining a VA better than 20/40, 20/65, and 20/200 in the study eye 1 year after diagnosis was 16% (95% CI, 4%–34%), 30% (95% CI, 16%–46%), and 42% (95% CI, 30%–54%) of eyes that had this VA level at diagnosis, respectively. At 3 years, 4 study eyes had a VA better than 20/200. The Kaplan-Meier estimate for not having low vision was 25% (95% CI, 14%–39%) at 1 year and 11% (95% CI, 4%–24%) at 2 years, and the proportion of eyes that were not blind was 54% (95% CI, 42%–66%) at 1 year, 26% (95% CI, 16%–37%) at 2 years, and 5% (95% CI, 0.5%–16%) at 5 years by WHO criteria (Fig 2F).

The Kaplan-Meier estimate for losing 6 or more lines of vision from baseline was 54% (95% CI, 44%–66%) by 1 year, 86% (95% CI, 76%–93%) by 2 years, and 96% (95% CI, 87%–99%) by 3 years after treatment.

A total of 78 person-years of VA better than 20/200 (median, 0.8 years per patient who initially had this level; range, 0.03–8.2 years), 34 person-years of VA better than 20/65 (median, 0.3 years; range, 0.05–5.5 years), and 19 person-years of VA better than 20/40 (median, 0.3 years; range, 0.09–4.2 years) in the study eye were saved. Likewise, 49 person-years without low vision (median, 0.6 years; range, 0.04–8.2 years) and 111 person-years without blindness (median, 1.0 year; range, 0.03–8.6 years) of the study eye were saved. In this series, 7 of 97 patients avoided low vision and 16 patients avoided blindness for at least 2 years in the treated eye.

Representative clinical outcomes after IBT are illustrated in Figure 3.

Factors Predicting Visual Loss

Mean VA over time, stratified by baseline variables, showed little difference according to age at diagnosis (Fig 4A), but it decreased with decreasing baseline VA (Fig 4B) and with increasing tumor height (Fig 4C). The LBD was not closely associated with mean VA (Fig 4D). Location of the anterior tumor margin in the ciliary body as compared with location behind the ora serrata (Fig 4E) and a distance of more than 8 mm from the posterior tumor margin to the FAZ were associated with higher mean VA over time than shorter distances (Fig 4F).

By univariate, single-failure Cox regression analysis, tumor height was a significant risk factor for developing low vision and blindness in the study eye (HR, 1.38 and 1.31 for each millimeter of increase; $P = 0.001$). Age at diagnosis (HR, 1.04 for each year

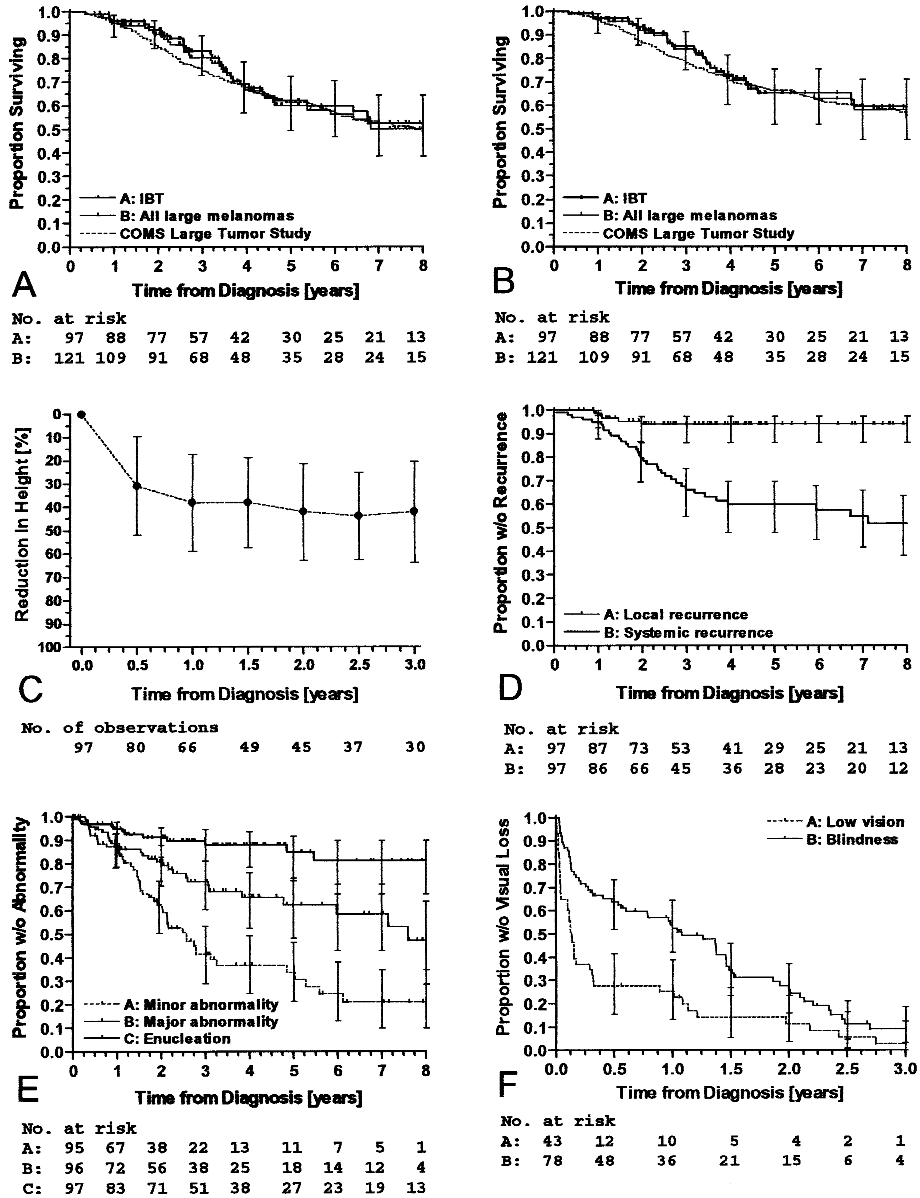

Figure 2. **A,** Kaplan-Meier plot of all-cause survival for 97 patients with large uveal melanoma by Collaborative Ocular Melanoma Study (COMS) criteria who were managed with iodine brachytherapy (IBT). Survival for the entire series of 121 patients with large uveal melanoma is seen, and that for patients who underwent enucleation without prior irradiation in the COMS Large Tumor Study[16] are shown for comparison. **B,** Corresponding plot of melanoma-specific survival, based on histopathologically confirmed or suspected melanoma metastasis. **C,** Percentage reduction in tumor height after IBT. **D,** Kaplan-Meier plot of time to local and systemic tumor recurrence. **E,** Kaplan-Meier plot of time to development of minor and major cosmetic abnormality, and to secondary enucleation. **F,** Kaplan-Meier plot of time to first occurrence of low vision and blindness in the study eye by World Health Organization definitions. Bars indicate 95% confidence intervals (in **A** and **B**, for patients managed with IBT).

increase; $P = 0.008$) was also a significant risk factor for low vision (Table 2).

By multiple-failure analysis, which allows repeated loss and recovery of vision (Table 2), location of the anterior tumor margin behind the ora serrata became a significant predictor of blindness (HR, 1.68; $P = 0.011$), and distance of the posterior tumor margin from the FAZ became a predictor of both low vision and blindness (HR, 0.94 for each millimeter of increase; $P = 0.009$ and 0.010, respectively) in the study eye. The LBD was not significantly associated with visual loss in any analysis.

Bivariate regression identified tumor height, on one hand, and location of the anterior tumor margin behind the ora serrata and short distance from posterior tumor margin to the FAZ, on the other hand, as independent predictors of both low vision and blindness in the study eye. Model 1, which combined height with location of the anterior tumor margin, better predicted develop-

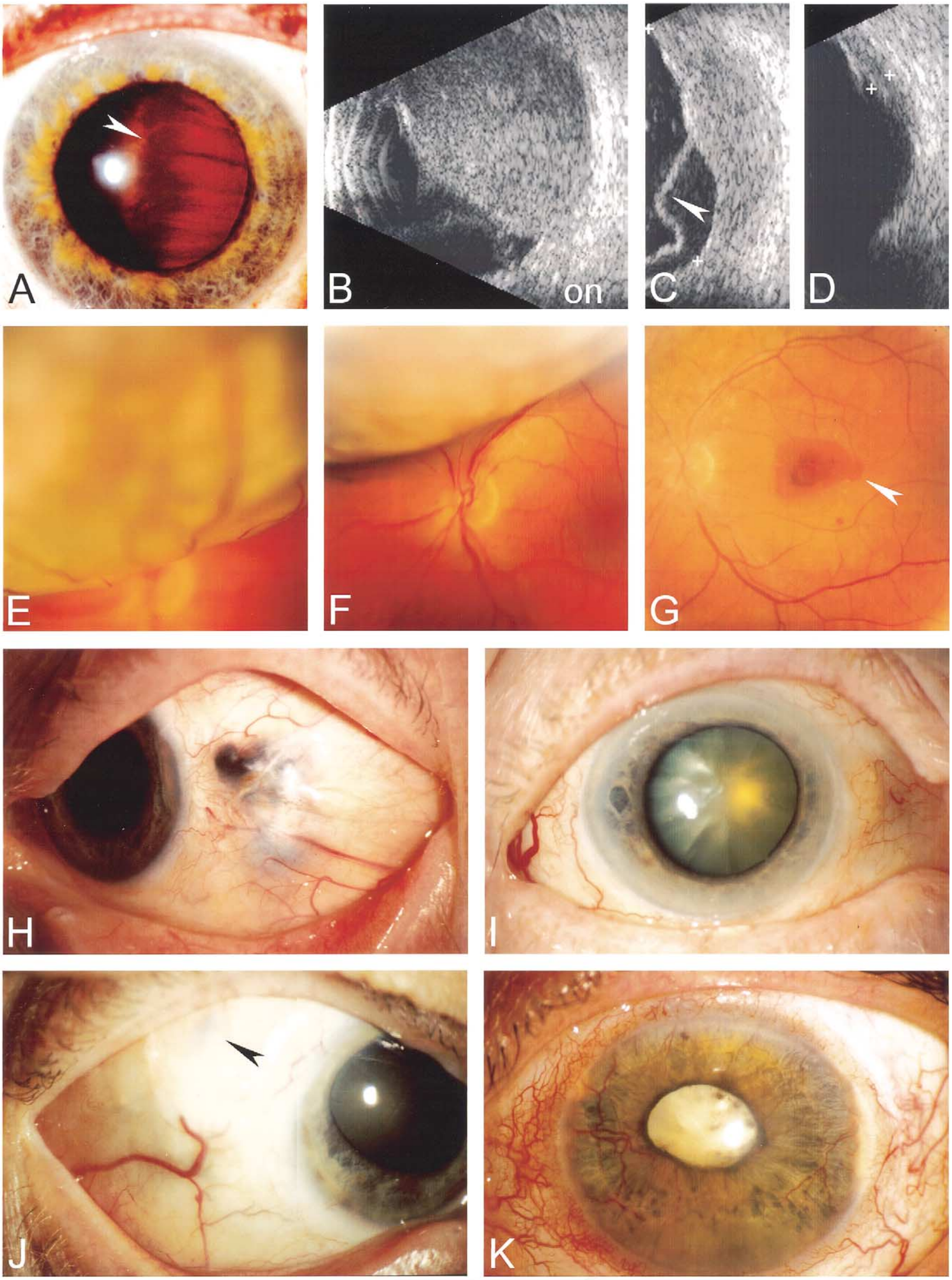
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Figure 3. Clinical outcomes after iodine brachytherapy (IBT) for large uveal melanoma by Collaborative Ocular Melanoma Study (COMS) criteria. **A,** Large ciliochoroidal melanoma (height, 16.8 mm; largest basal diameter [LBD], 21.4 mm) in a 40-year-old man with concurrent metastasis. The temporally located tumor pushes pars plana and ora serrata (arrowhead) against the lens. **B,** Posteriorly, the tumor extends close to the optic nerve (on) by ultrasonography. **C,** Six months after IBT, tumor height has reduced to 6.9 mm but exudative retinal detachment (arrowhead) persists. **D,** Eighteen months after IBT, the retina is attached and tumor height is 1.9 mm. Vision improved from hand movements to counting fingers. **E,** Large choroidal melanoma (height, 9.9 mm; LBD, 22.5 mm) in a 39-year-old man with 20/25 vision. **F,** Three months after IBT, tumor height is 8 mm. **G,** He retained better than 20/65 vision for 2.1 years, by which time maculopathy developed (arrowhead) and tumor height was 4 mm. **H,** Marginal extrascleral recurrence 2.2 years after IBT from incomplete coverage of the anterior tumor margin of a large ciliochoroidal melanoma (height, 9.0 mm; LBD, 20.5 mm) in a 64-year-old man. **I,** Radiation cataract as a minor cosmetic abnormality 12 months after IBT of a large ciliochoroidal melanoma (height, 10 mm; LBD, 25 mm). **J,** Scleral thinning (arrowhead) as a minor cosmetic abnormality 30 months after IBT of a large ciliochoroidal melanoma (height, 10 mm; LBD, 25 mm). **K,** Persistent redness from neovascular glaucoma, iris neovascularization, and cataract as a major cosmetic abnormality 36 months after IBT for a large peripapillary melanoma (height, 11.5 mm; LBD, 11.9 mm). At last review 4 years later, the eye was blind but asymptomatic.

ment of low vision in the study eye by single-failure analysis than Model 2, which included distance to the FAZ, and it was comparable with the latter by multiple-failure analysis (Table 2; difference in −2 log likelihood for single- and multiple-failure analysis, 7.42 and 2.92; chi-square test, 1 df, $P = 0.0065$ and 0.087, respectively). The two models were comparable in predicting development of blindness ($P = 0.084$ and 0.061, respectively), but in a trivariate model, location of the anterior tumor margin became the predominant predictor of blindness as well, together with tumor height (Table 2).

Discussion

The 5-year all-cause and melanoma-specific mortality of patients who had a large uveal melanoma managed with IBT seemed to be comparable with that reported for enucleation only in the COMS Large Tumor Study[16] and in other studies with predominantly large uveal melanomas.[32,33] This is despite that, unlike COMS, we did not exclude patients who had a history of major illness, other cancer, or metastases at diagnosis. The median LBD was slightly smaller than in the COMS Large Tumor Study,[16] but median tumor height was comparatively larger, indicating no major overall difference in tumor volume. We used criteria similar to COMS when coding the cause of death, and approximately 65% of melanoma deaths were histopathologically confirmed in both series.[16] Taken together, the data support the concept that primary enucleation provides little survival benefit for the patient as compared with globe-preserving therapy.[5]

Data on local tumor control after IBT in the COMS Medium Tumor Study have not yet been released.[15] In several observational series, the 5-year Kaplan-Meier estimate for recurrence after IBT in patients with mainly medium-sized tumors has ranged from 8% to 20%.[20,34,35] The 5-year 6% Kaplan-Meier estimate for having recurrence compares favorably with these data, regardless of the fact that, if indicated to reduce radiation to the optic disc and macula or to cover a very large tumor, we did not require an absolute 2-mm safety margin around the tumor. Our plaques differed from the COMS plaques in that they do not have a collimating rim, and coverage of the tumor with smaller safety margins was considered sufficient because of lateral scatter of radiation.[36] We also used a prescription dose lower than the 85 Gy required by COMS to treat very thick tumors and allowed the dose rate to vary more widely.[15]

In several eyes, treatment-related complications eventually prevented view to the fundus, and the possibility must be considered that we might have missed some recurrences. Secondary enucleation was offered when visibility was lost, but most patients declined further therapy in the absence of evidence of recurrence unless the eye was irritated and enucleation promised symptomatic relief. We have so far not observed delayed recurrences, although these patients are reviewed closely both by standard and high-frequency ultrasonography and by periodic CT and MRI, and the longest follow-up times exceed 8 years. Our data suggest that large tumor size does not predict more frequent local recurrence after IBT as compared with medium-sized tumors.[20,34,35] This conclusion must be revisited, however, when follow-up accumulates.

Local recurrence has been associated with a high risk of metastasis,[7,25,37] but it is not known whether this is because tumor cells disseminate from the recurrence or because of greater malignancy of tumors that recur. It is possible that some large tumors, which would have recurred locally after IBT, caused metastatic death instead before a local recurrence was observed.

In approximately 80% of patients, the eye was successfully conserved at 8 years, and 33% of patients had no cosmetic problem at 5 years. However, treatment-related complications both from irradiation and the presence of a large dying tumor, such as neovascular glaucoma, exudative retinal detachment, and vitreous hemorrhage were frequent. These often caused irritation and called for symptomatic therapy at some point during follow-up, and by 5 years, 38% of treated eyes had a major cosmetic abnormality.

Chances of developing such problems is probably one major reason why enucleation often is recommended as the primary therapy for large uveal melanoma. This used to be our practice, too, but we have since learned that, given the option, most of our patients elect IBT despite these unfavorable odds. Ocular oncologists in general pay more and more attention to the patient's own objectives when discussing treatment.[4,38] A limitation of our study is that we failed to address prospectively the quality-of-life aspects after treatment, and such an analysis was no longer feasible without introducing bias because many patients had died.

The conservation of visual acuity was less successful, which is not surprising because tumor size is one of the strongest risk factors for loss of vision in irradiated eyes.[10,17] In some patients, however, useful vision was not

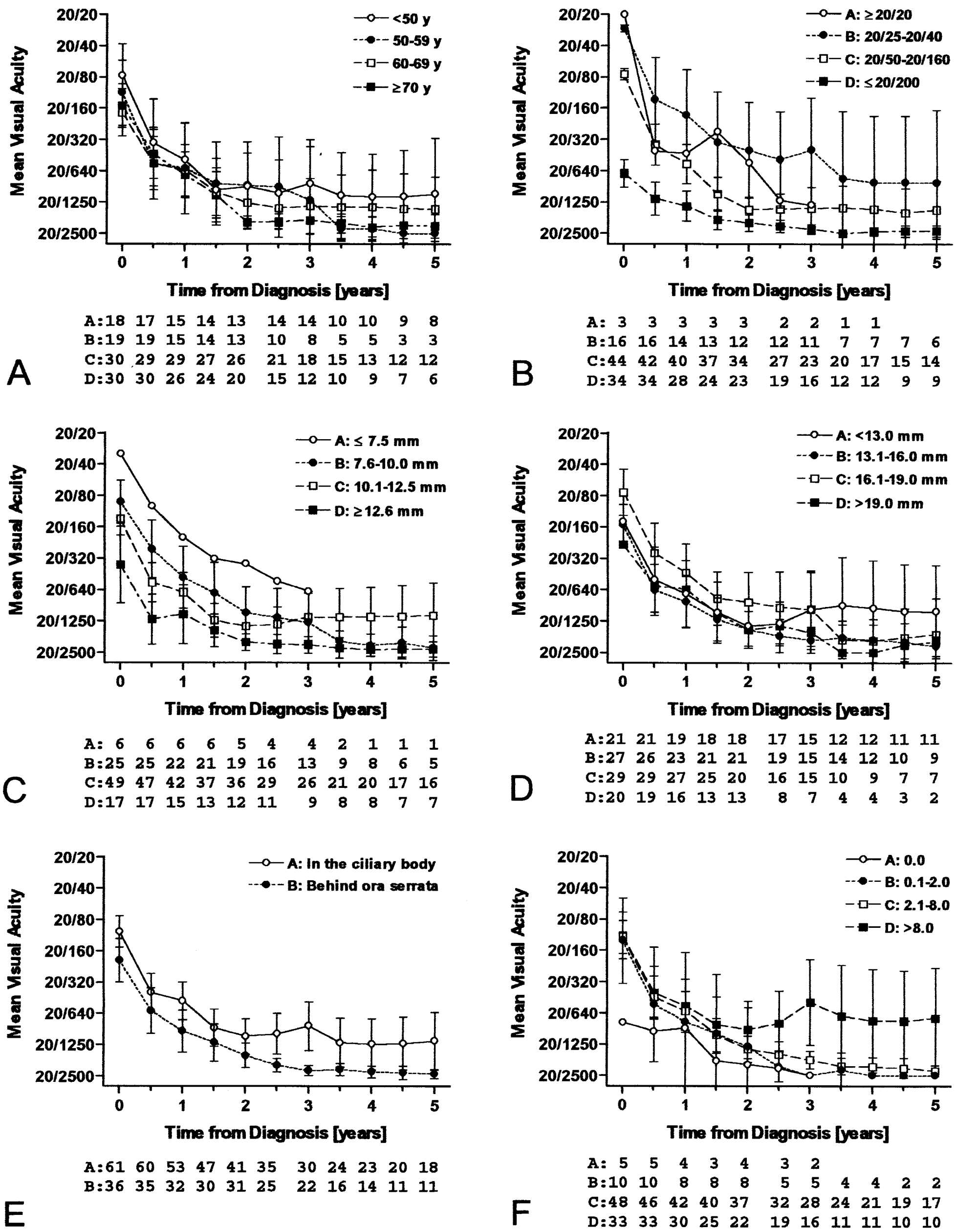

Figure 4. Mean visual acuity for 97 patients with a large uveal melanoma by Collaborative Ocular Melanoma Study (COMS) criteria who were managed with iodine brachytherapy (IBT), stratified by **A,** age; **B,** baseline visual acuity; **C,** tumor height; **D,** largest basal tumor diameter; **E,** location of the anterior tumor margin; and **F,** distance from the posterior tumor margin to the foveal avascular zone (FAZ). Bars show 95% confidence intervals.

lost, and a few maintained even reading vision in the tumor eye for the rest of their lives. In our study, the number needed to treat was 14 patients to allow one patient to retain 20/65 vision or better in the treated eye for at least 2 years, and the number needed to treat was 6 to avoid blindness of the treated eye in one patient for at least 2 years. It is impossible to predict which patients will enjoy such benefits after IBT, but both stratification by potential prognostic factors and Cox multiple regression analysis suggested that location of the anterior tumor margin in the ciliary body in addition to decreasing tumor height was the factor most consistently associated with smaller risk of both low vision and blindness in the treated eye.

Although the long-term prognosis of preserving vision in an irradiated eye is poor for most patients with a large uveal melanoma, even short-term conservation of vision in the tumor eye seems preferable to many patients as compared with immediate blindness from enucleation. Because benefit from enucleation as regards avoiding potential treatment-related complications is smallest for patients in whom metastases develop early and who die soon after diagnosis, it may be useful to remember that large uveal melanomas with

Table 2. Cox Proportional Hazards Regression of Time to Low Vision (20/70 or Worse) and Blindness (Loss of 20/400) in the Study Eye, Based on Single and Multiple Failure Data Sets

Variable	Single Failure				Repeated Failure			
	Coefficient (Standard Error)	*Wald Chi-square*	*P*	*Hazard Ratio (95% Confidence Interval)*	*Coefficient (Standard Error)*	*Wald Chi-square*	*P*	*Hazard Ratio (95% Confidence Interval)*
Univariate analysis								
Low vision								
Age*	0.036 (0.014)	6.97	0.008	1.04 (1.01–1.07)	0.022 (0.007)	8.06	0.005	1.02 (1.01–1.04)
Tumor height†	0.323 (0.096)	11.4	0.001	1.38 (1.14–1.67)	0.263 (0.066)	15.8	<0.001	1.30 (1.14–1.48)
Largest basal diameter†	−0.026 (0.048)	0.30	0.58	0.97 (0.89–1.07)	−0.002 (0.031)	0.005	0.95	1.00 (0.94–1.06)
Anterior tumor margin‡	0.500 (0.354)	1.99	0.16	1.65 (0.82–3.30)	0.370 (0.213)	3.03	0.082	1.45 (0.95–2.20)
Distance to FAZ†	−0.044 (0.034)	1.64	0.20	0.96 (0.90–1.02)	−0.065 (0.025)	6.76	0.009	0.94 (0.89–0.98)
Blindness								
Age*	0.014 (0.011)	1.72	0.19	1.01 (0.99–1.04)	0.017 (0.008)	4.33	0.038	1.02 (1.00–1.03)
Tumor height†	0.272 (0.081)	11.4	0.001	1.31 (1.12–1.54)	0.197 (0.059)	11.2	0.001	1.22 (1.08–1.37)
Largest basal diameter†	−0.028 (0.036)	0.58	0.45	0.97 (0.91–1.04)	0.019 (0.027)	0.49	0.49	1.02 (0.97–1.07)
Anterior tumor margin‡	0.377 (0.268)	1.99	0.16	1.46 (0.86–2.46)	0.518 (0.203)	6.50	0.011	1.68 (1.13–2.50)
Distance to FAZ†	−0.036 (0.030)	1.42	0.23	0.96 (0.91–1.02)	−0.065 (0.025)	6.55	0.010	0.94 (0.89–0.98)
Bivariate analysis								
Low vision								
Model 1	*−2 log likelihood = 214.05*				*−2 log likelihood = 351.18*			
Tumor height†	0.427 (0.101)	17.8	<0.001	1.53 (1.26–1.87)	0.310 (0.066)	22.0	<0.001	1.36 (1.20–1.55)
Anterior tumor margin‡	1.237 (0.408)	9.18	0.002	3.45 (1.55–7.67)	0.672 (0.206)	10.0	0.001	1.96 (1.31–2.93)
Model 2	*−2 log likelihood = 221.47*				*−2 log likelihood = 354.10*			
Tumor height†	0.315 (0.096)	10.8	0.001	1.37 (1.14–1.65)	0.239 (0.064)	14.1	<0.001	1.27 (1.12–1.44)
Distance to FAZ†	−0.035 (0.035)	1.02	0.31	0.97 (0.90–1.03)	−0.049 (0.018)	7.08	0.008	0.95 (0.92–0.99)
Blindness								
Model 1	*−2 log likelihood = 418.67*				*−2 log likelihood = 614.12*			
Tumor height†	0.333 (0.089)	14.1	<0.001	1.39 (1.17–1.66)	0.266 (0.065)	16.6	<0.001	1.30 (1.15–1.48)
Anterior tumor margin‡	0.689 (0.285)	5.86	0.016	1.99 (1.14–3.48)	0.796 (0.210)	14.3	<0.001	2.22 (1.47–3.35)
Model 2	*−2 log likelihood = 415.68*				*−2 log likelihood = 610.62*			
Tumor height†	0.264 (0.081)	10.5	0.001	1.30 (1.11–1.52)	0.189 (0.060)	9.99	0.002	1.21 (1.07–1.36)
Distance to FAZ†	−0.022 (0.300)	0.59	0.44	0.98 (0.92–1.03)	−0.060 (0.025)	5.80	0.016	0.94 (0.90–0.99)
Trivariate analysis								
Blindness								
Model 3	*−2 log likelihood = 410.62*				*−2 log likelihood = 604.86*			
Tumor height†	0.338 (0.091)	13.9	<0.001	1.40 (1.17–1.68)	0.253 (0.062)	16.6	<0.001	1.29 (1.14–1.45)
Anterior tumor margin‡	0.763 (0.337)	5.11	0.024	2.14 (1.11–4.15)	0.670 (0.243)	7.62	0.006	1.95 (1.21–3.14)
Distance to FAZ†	0.014 (0.034)	0.17	0.68	1.01 (0.95–1.08)	−0.027 (0.027)	1.04	0.31	0.97 (0.92–1.03)

FAZ = foveal avascular zone.
*Continuous variable, years.
†Continuous variable, mm.
‡Categories: 0 = in the ciliary body, 1 = behind the ora serrata.

ciliary body involvement are associated with short survival,[39] whereas this subgroup of patients had a better than average chance of preserving vision after IBT in our study.

We believe that most large melanomas can be managed safely with IBT in patients who are unwilling to undergo primary enucleation. In certain eyes, other conservative treatments may provide better chances of saving vision, however. In particular, tumors that are large because they are more than 10 mm in height but less than 16 mm by LBD may be managed better with transscleral local resection, especially when limited to the choroid.[11] Finally, patients with major complications from their tumor at diagnosis may still need primary enucleation.

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